

Education			
Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%

Experience

Google Summer of Code 2025 Contributor | P4 Language Consortium, Stanford, CA (May 2025 – Sept 2025)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 P4, C++, Python, P4Runtime, Docker, Intel Tofino

- Architected a switch-native framework compiling ML decision trees directly into P4 data planes using Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs, working around hardware ASIC constraints on Intel Tofino silicon.
- Built a C++/P4Runtime data-path testing pipeline spanning Tofino hardware and Mininet software targets, improving deployment reliability by 30%.
- Optimized low-level SID recirculation logic to hit 10 Gbps line-rate throughput with up to 5x lower inference latency than control-plane inference.

Open Source Contributions

- Project-HAMi/HAMi:** #2292 fix(device): use len(devices) for GenReason in kunlun and awsneuron; #2307 fix(device-plugin): release node lock when bind-phase patch fails; #2259 fix(scheduler): handle ResourceQuota tombstones.
- WasmEdge/WasmEdge:** #4470 fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch; #4466 feat(docker): add standalone alpine-base image build pipeline.
- facebook/bpfilter:** #507 build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

ZNIC v1 | C, Linux Kernel, PCI, DMA, NAPI, QEMU, DPDK (rte_ethdev) (Project Link)

Objective: Developed a clean-room Linux Ethernet driver with robust fault recovery and a user-space DPDK forwarding path.

Approach: Implemented a C driver featuring PCI MMIO, DMA rings, interrupt-driven NAPI, QEMU automation, and fault injection.

Impact: Maintained zero packet loss across 646,400 packets, proving stability through 1,000 interface cycles and reset-under-traffic tests.

Los-Tecnicos | Go, Rust, Soroban, Raspberry Pi, ESP32, TypeScript (Project Link)

Objective: Meter and tokenize renewable-energy production from a physical sensor mesh rather than from self-reported data.

Approach: Built a Raspberry Pi/ESP32 sensor mesh and the Go backend that ingests it; wrote 4 Soroban smart contracts and drove software/hardware bring-up end to end.

Impact: Won Asia-Pacific Web3 bounties and was recognized by the Government of India.

Do-It | Go, SSE, HTTP, Raspberry Pi, OpenWrt, Local-first (Project Link)

Objective: Run a private device-to-device sync service directly on constrained home hardware with no cloud dependency.

Approach: Packaged a single static Go SSE/HTTP binary small enough to run on a Raspberry Pi, consumer routers, and repurposed Android handsets.

Impact: Delivers account-free synchronization entirely on local hardware, with no household data leaving the LAN.

Myprod | Go, Nomad, Traefik, WireGuard, Docker, bare-metal (Project Link)

Objective: Turn a fleet of heterogeneous bare-metal and VPS machines into one schedulable compute pool.

Approach: Built a Go control plane over Nomad scheduling and WireGuard mesh networking, with guarded node lifecycle operations and live per-node CPU/memory telemetry.

Impact: Runs mixed-architecture nodes (x86_64 and ARM64) under a single scheduler with idempotent DNS and ingress management.

Technical Skills

- Language & RTL:** Verilog, C, C++, Python, P4, Go
- High-Speed Systems & Interfaces:** PCIe, Ethernet, DPDK, DMA, BAR0 MMIO, NAPI, eBPF
- Simulation & Verification:** QEMU, Wireshark, Scapy, LLVM, MLIR, Fault Injection, CI/CD
- Tools & Platforms:** Linux Kernel (x86_64, ARM64), VS Code, Docker, Git

Publications

- R. Bharadwaj, S. Jha, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” 2026 ICCSC, Ghaziabad, India, Feb. 2026. DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

- 2x AKTU State TT Gold (2023, 2024); Robotics Olympiad Bronze, Thailand (2019); NASA, NPTEL & ISRO certifications